

TEACHING AI FOR FINANCE

SESSION 2

KERRY BACK, RICE UNIVERSITY
DEPARTMENT OF FINANCE, UNIVERSITY OF ZURICH

JOB POSTINGS: INTERNSHIPS AND EARLY CAREER

Phrase	Posting
“AI tools in your work or school projects”	PIMCO, 2027 summer intern, Munich
“hands-on experience with AI-powered tools from day one”	PIMCO, 2027 summer intern, Munich
“sound judgement in technology-enabled decision making”	BlackRock, 2027 quantitative masters intern, London
“proactive about using AI where possible”	RA Capital, junior associate, Boston
“AI-forward people”	RA Capital, healthcare associate, Boston
“AI tools to systematize workflows and models”	Ares, analyst, New York

Job postings, checked October 31, 2026

JOB POSTINGS: BUILDING TOOLS

Phrase	Posting
“leveraging tools like Claude Code”	D. E. Shaw, AI product analyst, Denver
“custom prompts, skill libraries, and automated workflows”	Schonfeld, AI strategy analyst, London
“It is a building role.”	Two Sigma, AI solutions developer, New York
“the judgment to figure out what to build in the first place”	Two Sigma, AI solutions developer, New York
“converting complex workflows into logical, step-by-step tasks”	T. Rowe Price, investment AI analyst, Baltimore and London

Job postings, checked October 31, 2026

JOB POSTINGS: INVESTMENT KNOWLEDGE

Phrase	Posting
“Translate complex investment problems into structured AI workflows”	T. Rowe Price, investment AI analyst
“Translate broad investment questions into testable data science workflows”	Wellington, data science analyst, Boston
“not just a technical interest in the data itself”	Wellington, data science analyst, Boston
“we will teach the domain to the right technical candidate”	Schonfeld, AI strategy analyst, London
“both a hands-on technical builder and a strategic partner to investment professionals”	Carlyle, AlpInvest engineer, New York

Job postings, checked October 31, 2026

JOB POSTINGS: CHECKING THE OUTPUT

Phrase	Posting
“limitations, confidence levels, and interpretation risk”	Wellington, data science analyst, Boston
“how these tools have evolved and where they break down”	Two Sigma, AI solutions developer, New York
“robust guardrails, compliance checks, and observability frameworks”	T. Rowe Price, investment AI analyst
“evaluation, monitoring, and human-in-the-loop patterns”	Carlyle, AlpInvest engineer, New York
“appropriate audit trails”	Schonfeld, AI strategy analyst, London
“problems without clean labels”	Point72, AI engineer, London (since closed; via AI Street)

Job postings, checked October 31, 2026

ELEMENTS OF AI FLUENCY

01

Using agents effectively

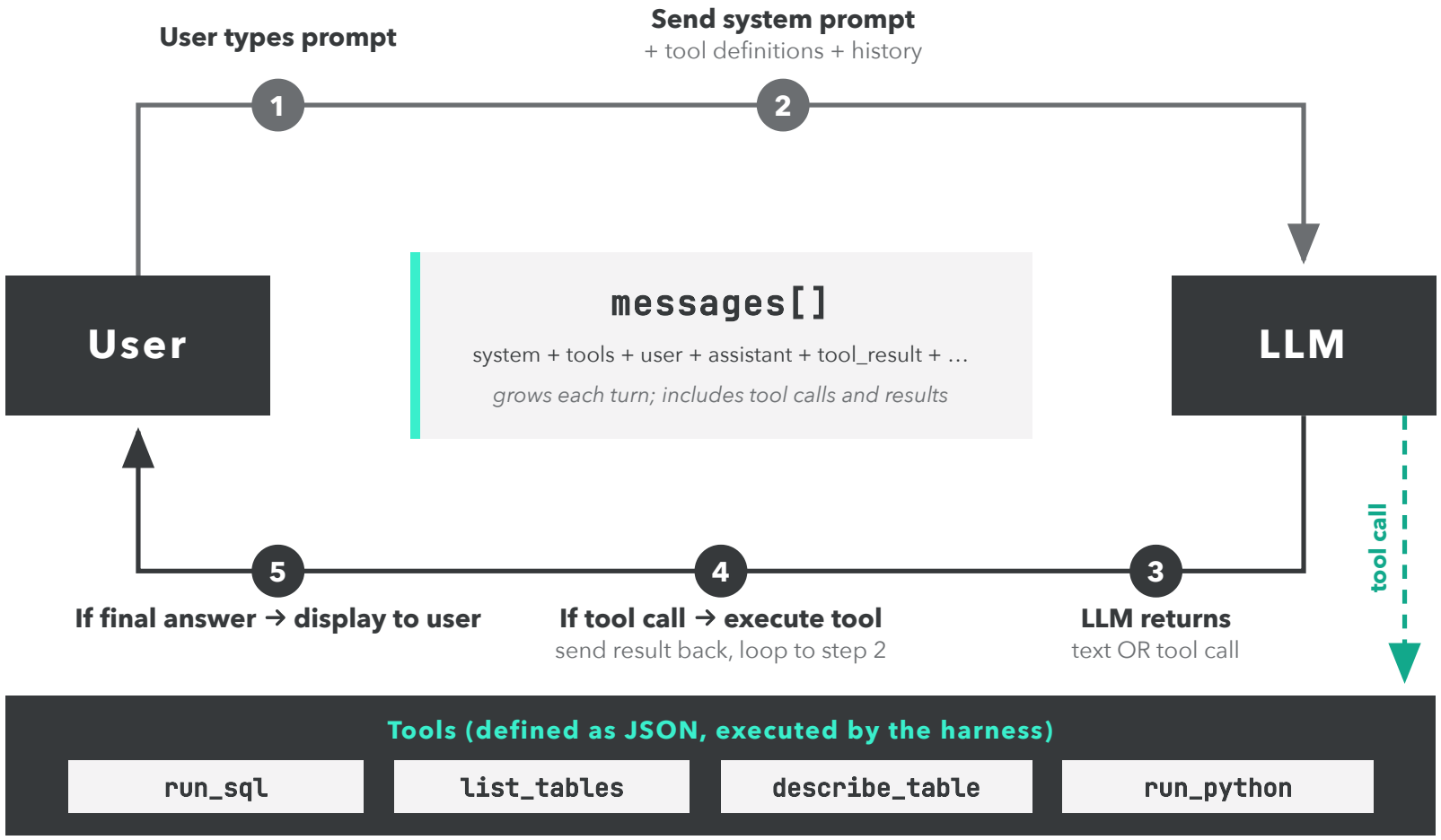
02

Creating custom tools

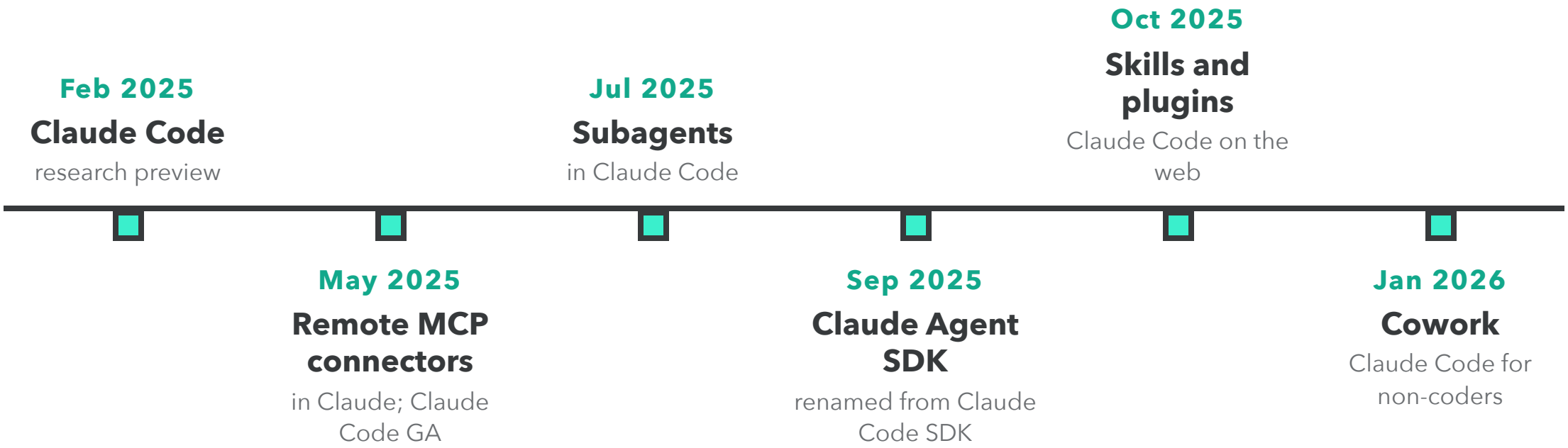
03

Understanding security and cost

ANATOMY OF AN AGENT



TIMELINE OF AGENT TOOLS



THIS IS NOT YOUR FATHER'S AI

October 2025

"They just don't work. They don't have enough intelligence … I feel like the industry is making too big of a jump and is trying to pretend like this is amazing, and it's not. It's slop."

January 2026

"I really am mostly programming in English now, a bit sheepishly telling the LLM what code to write… in words."

February–March 2026

"It is hard to communicate how much programming has changed due to AI in the last 2 months … Coding agents basically didn't work before December … I don't think I've typed like a line of code probably since December."

Andrej Karpathy: OpenAI co-founder, former Tesla AI chief, now at Anthropic

PROMPTING

Say what you want

Not what to do.

Ask for the best way (plan) first

Then tell it to proceed.

Ask AI to interrogate you

Let it decide what it needs to know.

Beware of tunnel vision

Ask: is there a better path?

Steelman

“What is the strongest argument against this?”

Numbers need code

“Compute this in Python.”

NEW STUDENT DELIVERABLES

- Software apps
- Skills
- Custom agents

VIBE CODING

- Students can submit software in addition to spreadsheets.
- Artifacts: single-page HTML apps. Claude can create and deploy.
- More complex software (with backend server). They can run locally or deploy in the cloud or company servers.
- Example: Tangency portfolio

AGENT SKILLS

- A skill is a text file of custom instructions for a particular task.
- It can be accompanied by code, including software programs that the agent launches and interacts with.
- Use Claude, ChatGPT, or Gemini with custom instructions and software tools in lieu of a custom agent.
- Interaction is via the usual prompt window.

PLUGINS

- Mechanism for distributing skills.
- In Claude desktop: Customize → Plugins → Add marketplace.
- Anthropic's plugin library: github.com/anthropics/claude-plugins-official
- kerryback-public plugin library: github.com/kerryback/kerryback-public

MCP SERVERS (CONNECTORS)

- An MCP server sits between an agent and an external service or database.
- It uses the external service's API and creates tools (functions) from them, which it describes in text form to a connected agent.
- The agent acquires the tools exposed by the server, so it can use the external service.
- Examples: Gmail, Microsoft 365, Slack, Workday, Salesforce, …

CUSTOM AGENTS

- Custom chatbots with custom tools making API calls.
- Easy to build with agent software development kits (SDKs): code that implements Claude Code or Codex or …
- Students can build them around class concepts.
- Example of an agent built with a two-line prompt: eia-agent.kerryback.com

EXAMPLE EXERCISES

1. Build a dashboard (app) that uses the EIA's API and shows trends in energy prices and volumes.
2. Connect Claude (or Codex, …) to an EIA MCP server (e.g., eia.kerryback.com/mcp). Create a skill that produces particular plots of energy prices and volumes when invoked.
3. Install the finance-data plugin from the kerryback-public library. Get 60 months of WMT returns using Yahoo Finance data and get MKT-RF and RF from French's data library for the same 60 months. Run the CAPM regression to estimate WMT's beta.

A SKILL FOR DCFS

- Instead of creating DCFs, students should create skills or custom agents for creating DCFs.
- Example: `/mydcf {ticker}` gets statements, 10-Ks, press releases, and call transcripts. It writes a page with each judgment call flagged, and a workbook with live formulas.

TEACHING STUDENTS TO CHECK AI WORK

Have AI explain its work

"Explain what you did step by step. Did you make any decisions that we have not discussed?"

Use another AI agent to check the work

A fresh conversation or a subagent (a second agent that the first agent starts), given your goals and the first agent's explanation.

Have AI build a spreadsheet when possible

Then check the formulas. Every number traces back to its inputs.

Run the same code on small examples

Ask AI to run its code on a few rows you can check by hand.

Most AI coding errors arise from misunderstandings and unstated assumptions — and from errors that people also make, for example, merging and filtering in the wrong order.

A SEPARATE COURSE

Part	Topics
Models and agents	How language models work; performance; agents; computer use
Working with AI	Doing work with AI; prompting; permissions
Standing instructions and connectors	CLAUDE.md; memory; SKILL.md; MCP; plugins
Building agents	How software works; API calls; a chatbot; an agent
Deploying agents	Running locally; Git and GitHub; deployment
Documents and local models	RAG; Gemini Notebook; local models
Agent security	Multi-agent architectures; risks; mitigation

MGMT 803, Gen AI for Business, Rice, four days, September 24–27, 2026. Graded on in-class exercises, screen shares, and participation. The class marketplace holds 28 plugins.



QUESTIONS?